CHAPTER 13

Visualizing Time Series and Other Functions of an Independent Variable

The preceding chapter discussed scatterplots, where we plot one quantitative variable against another. A special case arises when one of the two variables can be thought of as time, because time imposes additional structure on the data. Now the data points have an inherent order; we can arrange the points in order of increasing time and define a predecessor and successor for each data point. We frequently want to visualize this temporal order, and we do so with line graphs. Line graphs are not limited to time series, however. They are appropriate whenever one variable imposes an ordering on the data. This scenario arises also, for example, in a controlled experiment where a treatment variable is purposefully set to a range of different values. If we have multiple variables that depend on time, we can either draw separate line plots or we can draw a regular scatterplot and then draw lines to connect the neighboring points in time.

Individual Time Series

As a first demonstration of a time series, we will consider the pattern of monthly preprint submissions in biology. Preprints are scientific articles that researchers post online before formal peer review and publication in a scientific journal. The preprint server bioRxiv, which was founded in November 2013 specifically for researchers working in the biological sciences, has seen substantial growth in monthly submissions since. We can visualize this growth by making a form of scatterplot (Chapter 12) where we draw dots representing the number of submissions in each month (Figure 13-1).

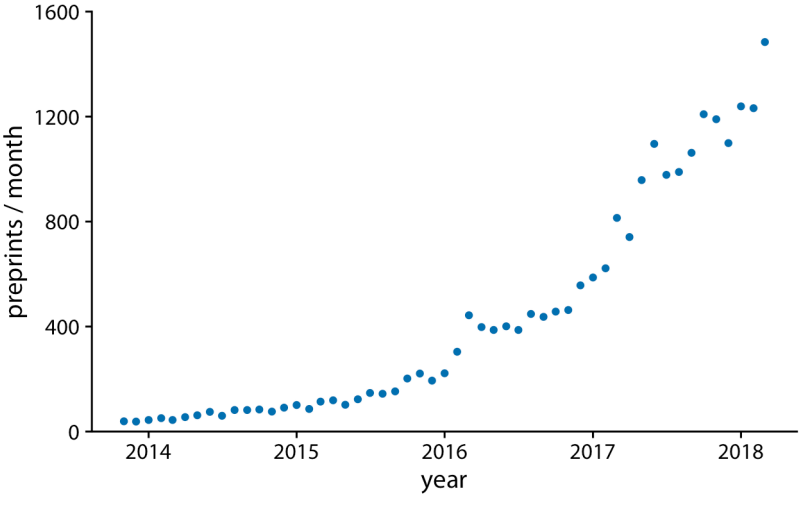


Figure 13-1. Monthly submissions to the preprint server bioRxiv, from its inception in November 2013 until April 2018. Each dot represents the number of submissions in one month. There has been a steady increase in submission volume throughout the entire 4.5-year period. Data source: Jordan Anaya, http://www.prepubmed.org.

There is an important difference, however, between Figure 13-1 and the scatterplots discussed in Chapter 12. In Figure 13-1, the dots are spaced evenly along the *x* axis, and there is a defined order among them. Each dot has exactly one left and one right neighbor (except the leftmost and rightmost points, which have only one neighbor each). We can visually emphasize this order by connecting neighboring points with lines (Figure 13-2). Such a plot is called a *line graph*.

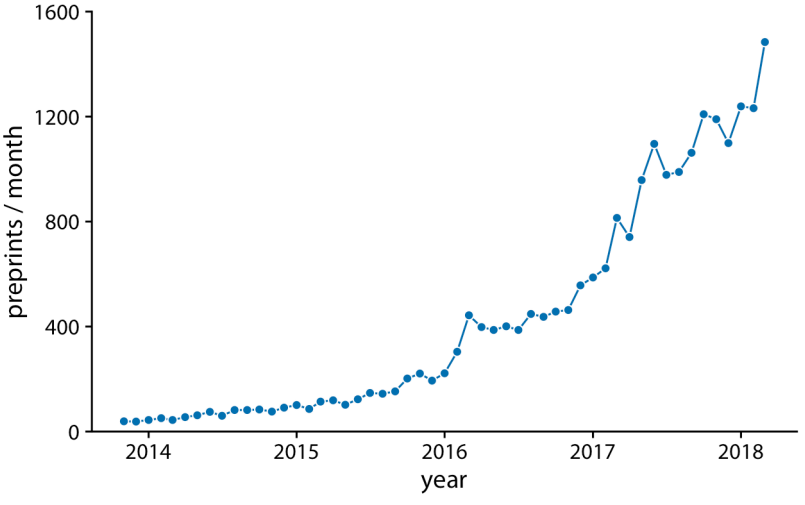


Figure 13-2. Monthly submissions to the preprint server bioRxiv, shown as dots connected by lines. The lines do not represent data and are only meant as a guide to the eye. By connecting the individual dots with lines, we emphasize that there is an order between the dots: each dot has exactly one neighbor that comes before it and one that comes after. Data source: Jordan Anaya, http://www.prepubmed.org.

Some people object to drawing lines between points because the lines do not represent observed data. In particular, if there are only a few observations spaced far apart, had observations been made at intermediate times they would probably not have fallen exactly onto the lines shown. Thus, in a sense, the lines correspond to made-up data. Yet they may help with perception when the points are spaced far apart or are unevenly spaced. We can somewhat resolve this dilemma by pointing it out in the figure caption, for example by writing "lines are meant as a guide to the eye" (see caption of Figure 13-2).

Using lines to represent time series is generally accepted practice, however, and frequently the dots are omitted altogether (Figure 13-3). Without dots, the figure places more emphasis on the overall trend in the data and less on individual observations. A figure without dots is also visually less busy. In general, the denser the time series, the less important it is to show individual observations with dots. For the preprint dataset shown here, I think omitting the dots is fine.

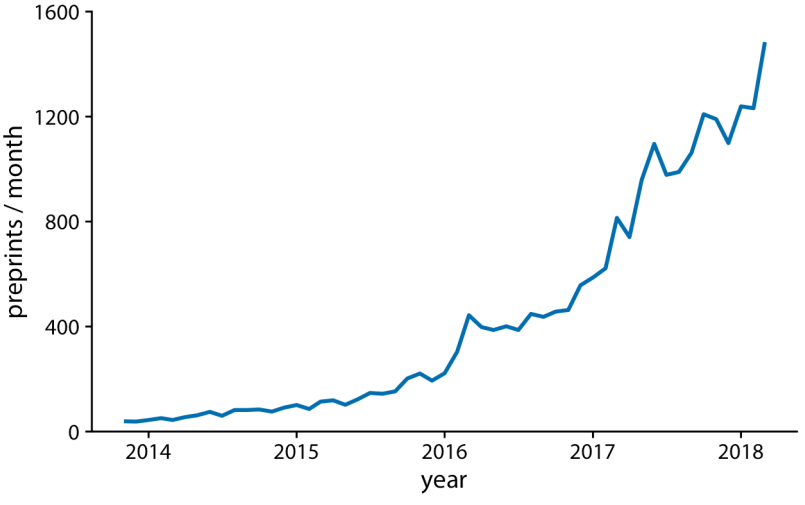


Figure 13-3. Monthly submissions to the preprint server bioRxiv, shown as a line graph without dots. Omitting the dots emphasizes the overall temporal trend while deemphasizing individual observations at specific time points. It is particularly useful when the time points are spaced very densely. Data source: Jordan Anaya, http://www.prepubmed.org.

We can also fill the area under the curve with a solid color (Figure 13-4). This choice further emphasizes the overarching trend in the data, because it visually separates the area above the curve from the area below. However, this visualization is only valid if the *y* axis starts at zero, so that the height of the shaded area at each time point represents the data value at that time point.

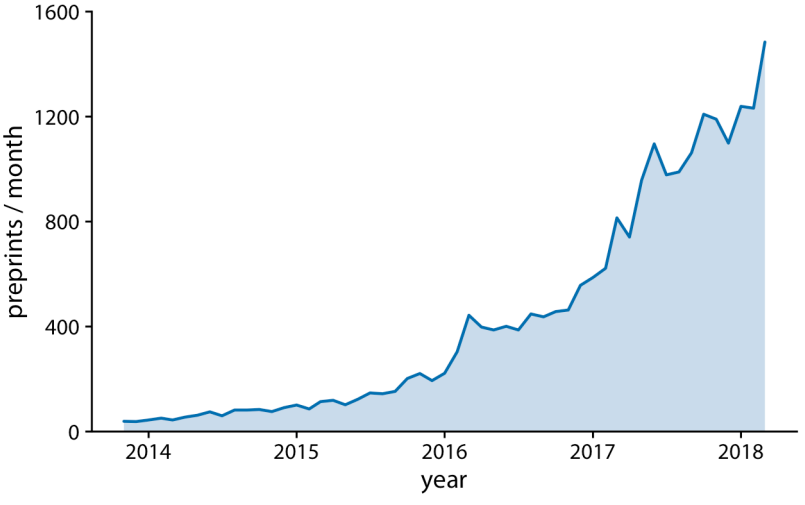


Figure 13-4. Monthly submissions to the preprint server bioRxiv, shown as a line graph with filled area underneath. By filling the area under the curve, we put even more emphasis on the overarching temporal trend than if we just draw a line (Figure 13-3). Data source: Jordan Anaya, http://www.prepubmed.org.

Multiple Time Series and Dose–Response Curves

We often have multiple time courses that we want to show at once. In this case, we have to be more careful in how we plot the data, because the figure can become confusing or difficult to read. For example, if we want to show the monthly submissions to multiple preprint servers, a scatterplot is not a good idea, because the individual time courses run into each other (Figure 13-5). Connecting the dots with lines alleviates this issue (Figure 13-6).

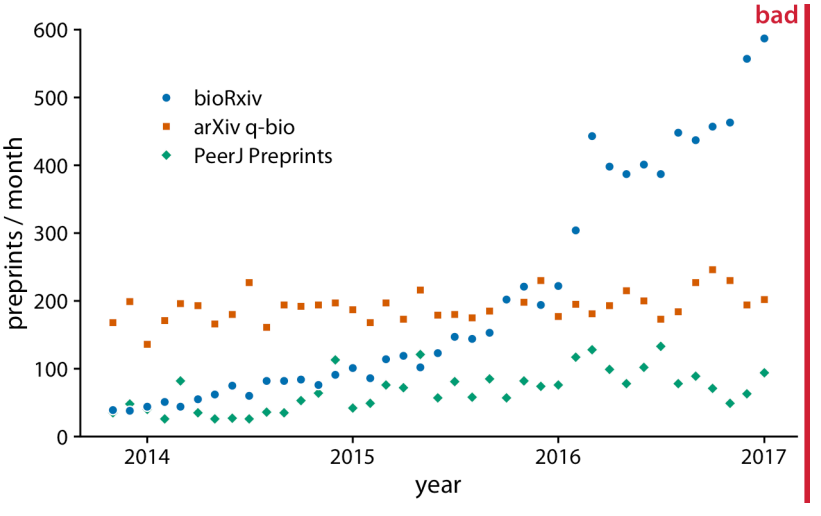


Figure 13-5. Monthly submissions to three preprint servers covering biomedical research: bioRxiv, the q-bio section of arXiv, and PeerJ Preprints. Each dot represents the number of submissions in one month to the respective preprint server. This figure is labeled "bad" because the three time courses visually interfere with each other and are difficult to read. Data source: Jordan Anaya, http://www.prepubmed.org.

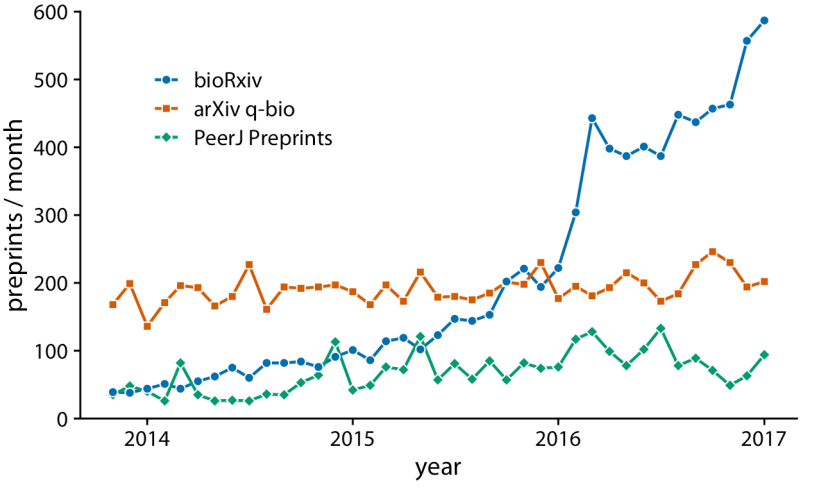


Figure 13-6. Monthly submissions to three preprint servers covering biomedical research. By connecting the dots in Figure 13-5 with lines, we help the viewer follow each individual time course. Data source: Jordan Anaya, http://www.prepubmed.org.

Figure 13-6 represents an acceptable visualization of the preprints dataset. However, the separate legend creates unnecessary cognitive load. We can reduce this cognitive load by labeling the lines directly (Figure 13-7). I have also eliminated the individual dots in this figure, for a result that is much more streamlined and easy to read than the original starting point, Figure 13-5.

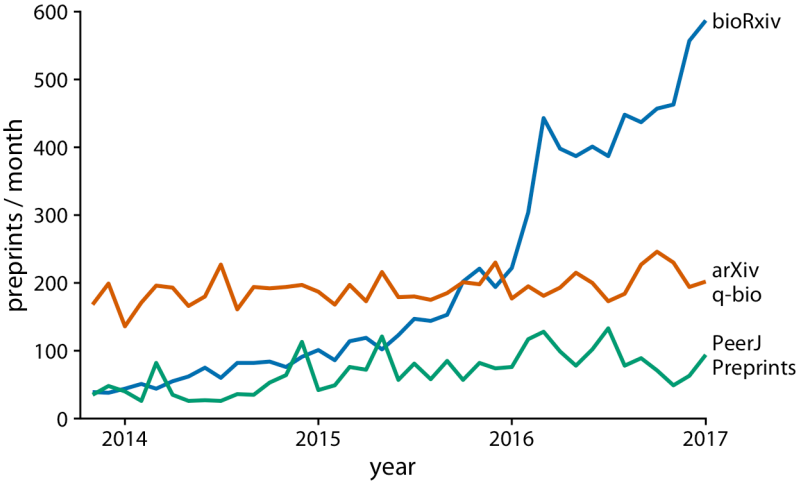


Figure 13-7. Monthly submissions to three preprint servers covering biomedical research. Directly labeling the lines instead of providing a legend reduces the cognitive load required to read the figure, and eliminating the legend removes the need for points of different shapes. This enables us to streamline Figure 13-6 further by eliminating the dots. Data source: Jordan Anaya, http://www.prepubmed.org.

Line graphs are not limited to time series. They are appropriate whenever the data points have a natural order that is reflected in the variable shown along the *x* axis, so that neighboring points can be connected with a line. This situation arises, for example, in dose–response curves, where we measure how changing some numerical parameter in an experiment (the dose) affects an outcome of interest (the response). Figure 13-8 shows a classic experiment of this type, measuring oat yield in response to increasing amounts of fertilization. The line graph visualization highlights how the dose–response curves have a similar shape for the three oat varieties considered but differ in the starting point in the absence of fertilization (i.e., some varieties have naturally higher yield than others).

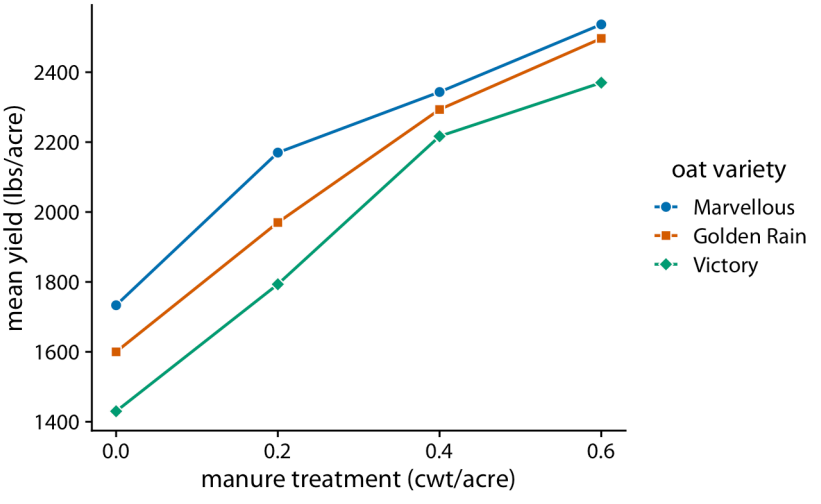


Figure 13-8. Dose–response curve showing the mean yield of oat varieties after fertilization with manure. The manure serves as a source of nitrogen, and oat yields generally increase as more nitrogen is available, regardless of variety. Here, manure application is measured in cwt (hundredweight) per acre. The hundredweight is an old imperial unit equal to 112 lbs or 50.8 kg. Data source: [Yates 1935].

Time Series of Two or More Response Variables

In the preceding examples we dealt with time courses of only a single response variable (e.g., preprint submissions per month or oat yield). It is not unusual, however, to have more than one response variable. Such situations arise commonly in macroeconomics. For example, we may be interested in the change in house prices from the previous 12 months as it relates to the unemployment rate. We may expect that house prices rise when the unemployment rate is low, and vice versa.

With the tools from the preceding sections, we can visualize such data as two separate line graphs stacked on top of each other (Figure 13-9). This plot directly shows the two variables of interest, and it is straightforward to interpret. However, because the two variables are shown as separate line graphs, drawing comparisons between them can be cumbersome. If we want to identify temporal regions when both variables move in the same or in opposite directions, we need to switch back and forth between the two graphs and compare the relative slopes of the two curves.

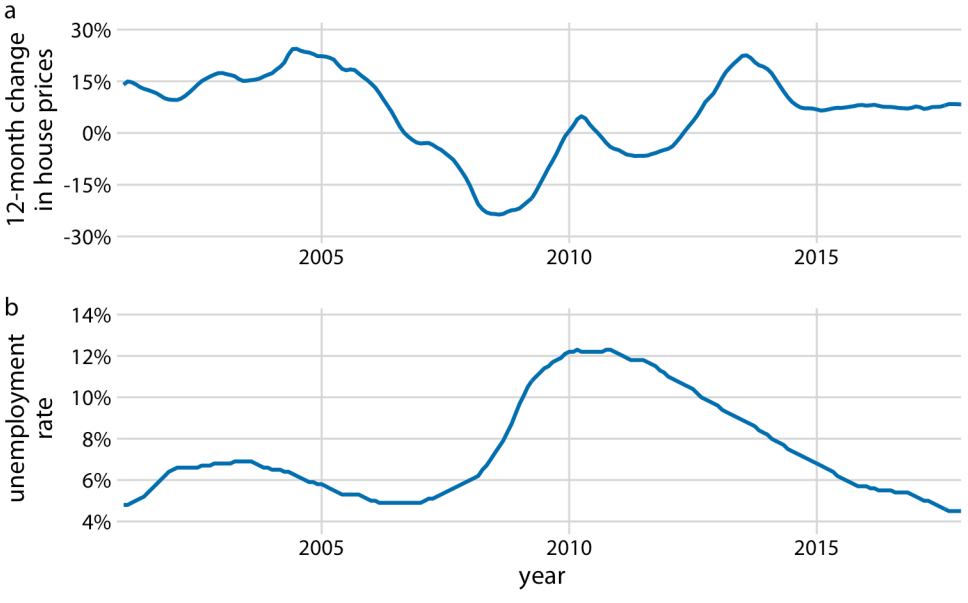


Figure 13-9. Twelve-month change in house prices (a) and unemployment rate (b) over time, from January 2001 through December 2017. Data sources: Freddie Mac House Prices Index, US Bureau of Labor Statistics.

As an alternative to showing two separate line graphs, we can plot the two variables against each other, drawing a path that leads from the earliest time point to the latest (Figure 13-10). Such a visualization is called a *connected scatterplot*, because we are technically making a scatterplot of the two variables against each other and then are connecting neighboring points. Physicists and engineers often call this a *phase portrait*, because in their disciplines it is commonly used to represent movement in phase space. We have previously encountered connected scatterplots in Chapter 3, where I plotted the daily temperature normals in Houston, TX, versus those in San Diego, CA (Figure 3-3).



Figure 13-10. Twelve-month change in house prices versus unemployment rate, from January 2001 through December 2017, shown as a connected scatterplot. Darker shades represent more recent months. The anticorrelation seen in Figure 13-9 between the change in house prices and the unemployment rate causes the connected scatterplot to form two counterclockwise circles. Original figure concept: Len Kiefer. Data sources: Freddie Mac House Price Index, US Bureau of Labor Statistics.

In a connected scatterplot, lines going in the direction from the lower left to the upper right represent correlated movement between the two variables (as one variable grows, so does the other), and lines going in the perpendicular direction, from the upper left to the lower right, represent anticorrelated movement (as one variable grows, the other shrinks). If the two variables have a somewhat cyclic relationship, we will see circles or spirals in the connected scatterplot. In Figure 13-10, we see one small circle from 2001 through 2005 and one large circle for the remainder of the time course.

When drawing a connected scatterplot, it is important that we indicate both the direction and the temporal scale of the data. Without such hints, the plot can turn into a meaningless scribble (Figure 13-11). In Figure 13-10 I used a gradual darkening of the color to indicate direction; alternatively, one could draw arrows along the path.

Is it better to use a connected scatterplot or two separate line graphs? Separate line graphs tend to be easier to read, but once people are used to connected scatterplots

they may be able to extract certain patterns (such as cyclical behavior with some irregularity) that can be difficult to spot in line graphs. In fact, to me the cyclical relationship between change in house prices and unemployment rate is hard to spot in Figure 13-9, but the counterclockwise spiral in Figure 13-10 reveals it. Research reports that readers are more likely to confuse order and direction in a connected scatterplot than in line graphs, and less likely to report correlation [Haroz, Kosara, and Franconeri 2016]. On the flip side, connected scatterplots seem to result in higher engagement, and thus such plots may be effective tools to draw readers into a story.

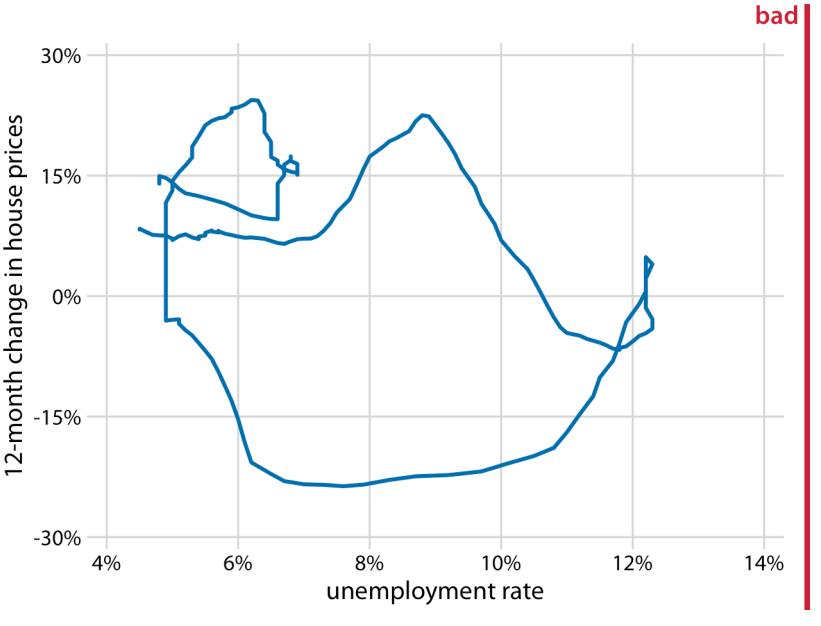


Figure 13-11. Twelve-month change in house prices versus unemployment rate, from January 2001 through December 2017. This figure is labeled "bad" because without the date markers and color shading of Figure 13-10, we can see neither the direction nor the speed of change in the data. Data sources: Freddie Mac House Prices Index, US Bureau of Labor Statistics.

Even though connected scatterplots can show only two variables at a time, we can also use them to visualize higher-dimensional datasets. The trick is to apply dimension reduction first (see Chapter 12). We can then draw a connected scatterplot in the dimension-reduced space. As an example of this approach, we will visualize a database of monthly observations of over 100 macroeconomic indicators, provided by the Federal Reserve Bank of St. Louis. We perform a principal components analysis (PCA) of all indicators and then draw a connected scatterplot of PC 2 versus PC 1 (Figure 13-12a) and versus PC 3 (Figure 13-12b).

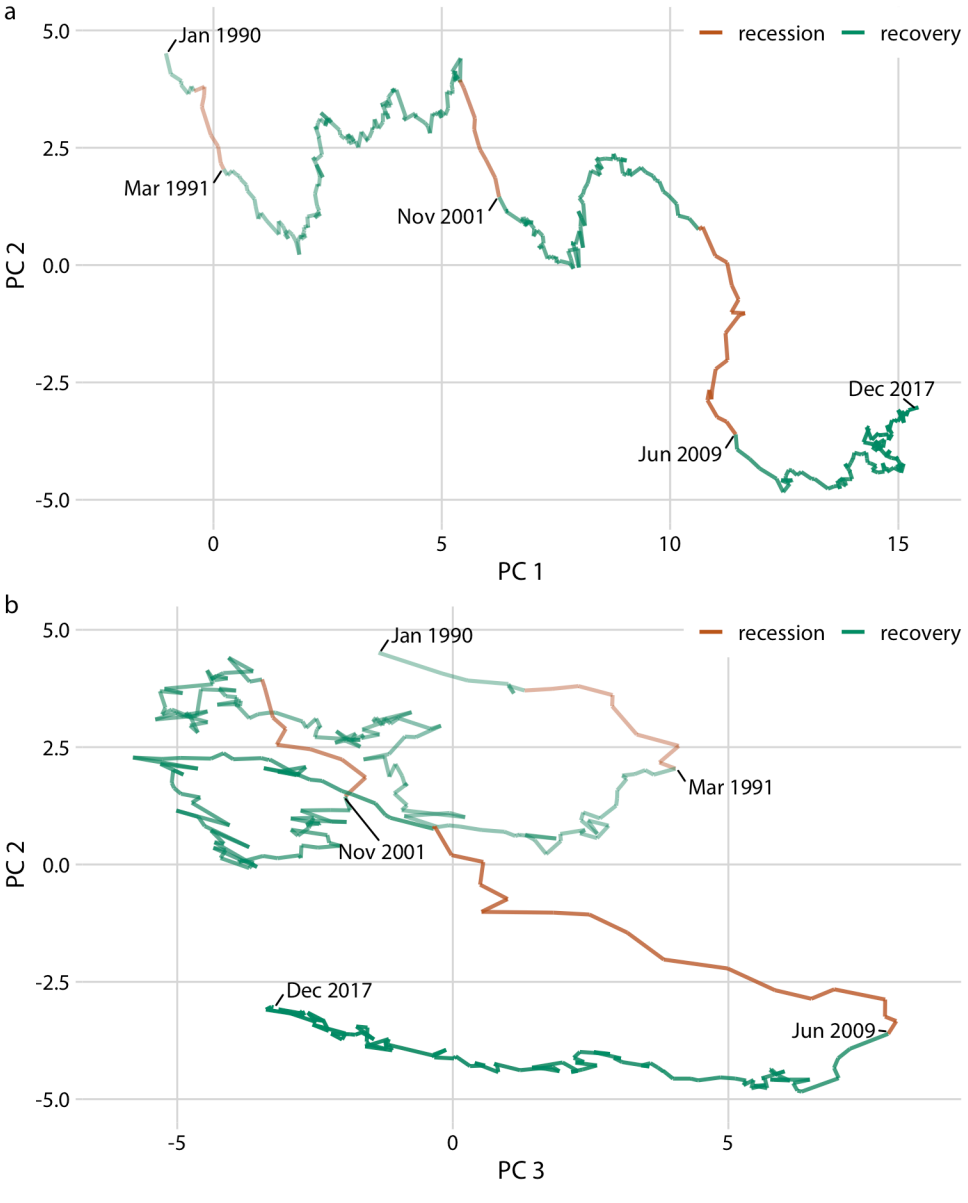


Figure 13-12. Visualizing a high-dimensional time series as a connected scatterplot in principal components space. The path indicates the joint movement of over 100 macroeconomic indicators from January 1990 to December 2017. Times of recession and recovery are indicated via color, and the endpoints of the three recessions (March 1991, November 2001, and June 2009) are also labeled. (a) PC 2 versus PC 1. (b) PC 2 versus PC 3. Data source: M. W. McCracken, St. Louis Fed.

Notably, Figure 13-12a looks almost like a regular line plot, with time running from left to right. This pattern is caused by a common feature of PCA: the first component often measures the overall size of the system. Here, PC 1 approximately measures the overall size of the economy, which rarely decreases over time.

By coloring the connected scatterplot by times of recession and recovery, we can see that recessions are associated with a drop in PC 2 whereas recoveries do not correspond to a specific feature in either PC 1 or PC 2 (Figure 13-12a). The recoveries do, however, seem to correspond to a drop in PC 3 (Figure 13-12b). Moreover, in the PC 2 versus PC 3 plot, we see that the line follows the shape of a clockwise spiral. This pattern emphasizes the cyclical nature of the economy, with recessions following recoveries and vice versa.

CHAPTER 14
Visualizing Trends

When making scatterplots (Chapter 12) or time series (Chapter 13), we are often more interested in the overarching trend of the data than in the specific detail of where each individual data point lies. By drawing the trend on top of or instead of the actual data points, usually in the form of a straight or curved line, we can create a visualization that helps the reader immediately see key features of the data. There are two fundamental approaches to determining a trend: we can either smooth the data by some method, such as a moving average, or we can fit a curve with a defined functional form and then draw the fitted curve. Once we have identified a trend in a dataset, it may also be useful to look specifically at deviations from the trend or to separate the data into multiple components, including the underlying trend, any existing cyclical components, and episodic components or random noise.

Smoothing

Let us consider a time series of the Dow Jones Industrial Average (Dow Jones for short), a stock market index representing the price of 30 large, publicly owned US companies. Specifically, we will look at the year 2009, right after the 2008 crash (Figure 14-1). During the tail end of the crash, in the first 3 months of the year 2009, the market lost over 2,400 points (~27%). Then it slowly recovered for the remainder of the year. How can we visualize these longer-term trends while deemphasizing the less important short-term fluctuations?



Figure 14-1. Daily closing values of the Dow Jones Industrial Average for the year 2009. Data source: Yahoo! Finance.

In statistical terms, we are looking for a way to *smooth* the stock market time series. The act of smoothing produces a function that captures key patterns in the data while removing irrelevant minor detail or noise. Financial analysts usually smooth stock market data by calculating *moving averages*. To generate a moving average, we take a time window, say the first 20 days in the time series, calculate the average price over these 20 days, then move the time window by one day, so it now spans the 2nd to 21st days. We then calculate the average over these 20 days, move the time window again, and so on. The result is a new time series consisting of a sequence of averaged prices.

To plot this sequence of moving averages, we need to decide which specific time point to associate with the average for each time window. Financial analysts often plot each average at the end of its respective time window. This choice results in curves that lag the original data (Figure 14-2a), with more severe lags corresponding to larger averaging time windows. Statisticians, on the other hand, plot the average at the center of the time window, which results in a curve that overlays perfectly on the original data (Figure 14-2b).

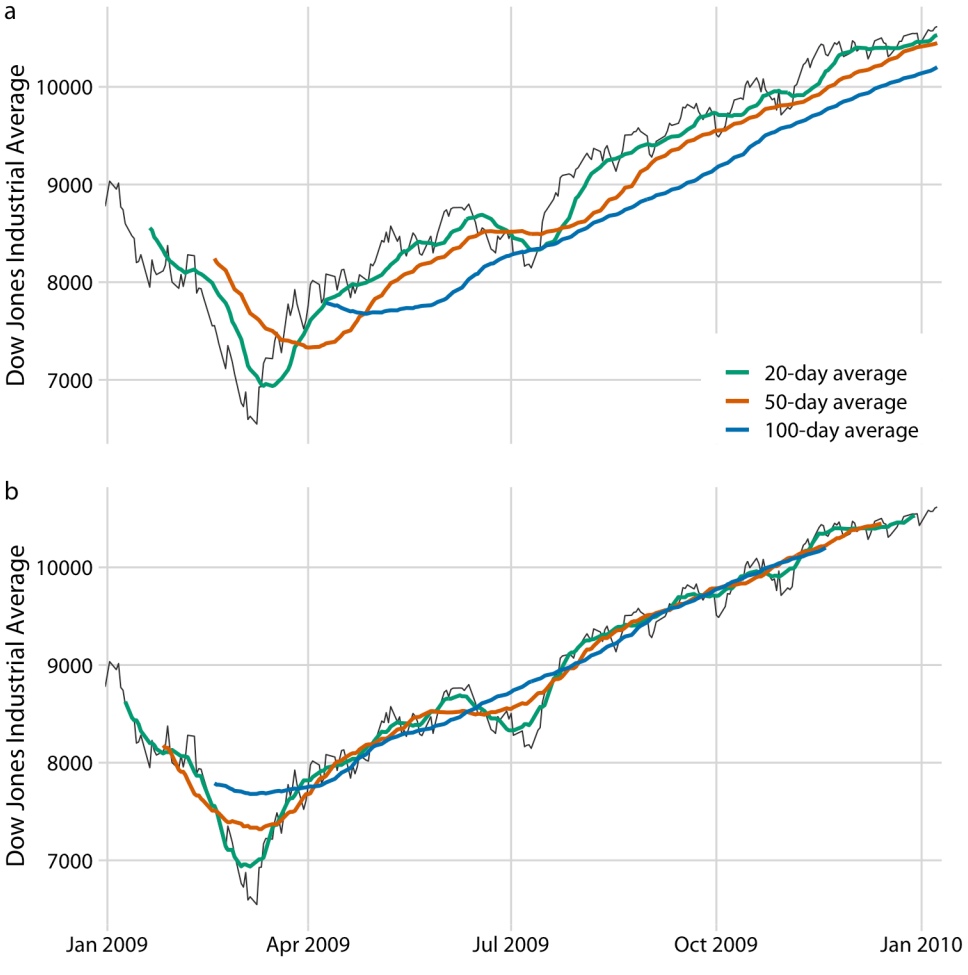


Figure 14-2. Daily closing values of the Dow Jones Industrial Average for the year 2009, shown together with their 20-day, 50-day, and 100-day moving averages. (a) The moving averages are plotted at the ends of the moving time windows. (b) The moving averages are plotted in the centers of the moving time windows. Data source: Yahoo! Finance.

Regardless of whether we plot the smoothed time series with or without lag, we can see that the length of the time window over which we average sets the scale of the fluctuations that remain visible in the smoothed curve. The 20-day moving average removes small, short-term spikes but otherwise follows the daily data closely. The 100-day moving average, on the other hand, removes even fairly substantial drops or spikes that play out over a time span of multiple weeks. For example, the massive drop to below 7,000 points in the first quarter of 2009 is not visible in the 100-day moving average, which replaces it with a gentle curve that doesn't dip much below

8,000 points (Figure 14-2). Similarly, the drop around July 2009 is completely invisible in the 100-day moving average.

The moving average is the most simplistic approach to smoothing, and it has some obvious limitations. First, it results in a smoothed curve that is shorter than the original curve (Figure 14-2). Parts are missing at either the beginning or the end or both. And the more the time series is smoothed (i.e., the larger the averaging window), the shorter the smoothed curve. Second, even with a large averaging window, a moving average is not necessarily that smooth. It may exhibit small bumps and wiggles even though larger-scale smoothing has been achieved (Figure 14-2). These wiggles are caused by individual data points that enter or exit the averaging window. Since all data points in the window are weighted equally, individual data points at the window boundaries can have a visible impact on the average.

Statisticians have developed numerous approaches to smoothing that alleviate the downsides of moving averages. These approaches are much more complex and computationally costly, but they are readily available in modern statistical computing environments. One widely used method is *locally estimated scatterplot smoothing* (LOESS) [Cleveland 1979], which fits low-degree polynomials to subsets of the data. Importantly, the points in the center of each subset are weighted more heavily than points at the boundaries, and this weighting scheme yields a much smoother result than we get from a weighted average. The LOESS curve shown in Figure 14-3 looks similar to the 100-day average in Figure 14-2, but this similarity should not be overinterpreted. The smoothness of a LOESS curve can be tuned by adjusting a parameter, and different parameter choices would have produced LOESS curves looking more like the 20-day or 50-day average.

Importantly, LOESS is not limited to time series. It can be applied to arbitrary scatterplots, as is apparent from its name, *locally estimated scatterplot smoothing*. For example, we can use LOESS to look for trends in the relationship between a car's fuel-tank capacity and its price (Figure 14-4). The LOESS line shows that tank capacity grows approximately linearly with price for cheap cars (below $20,000) but levels off for more expensive cars. Above a price of approximately $20,000, buying a more expensive car will not get you one with a larger fuel tank.

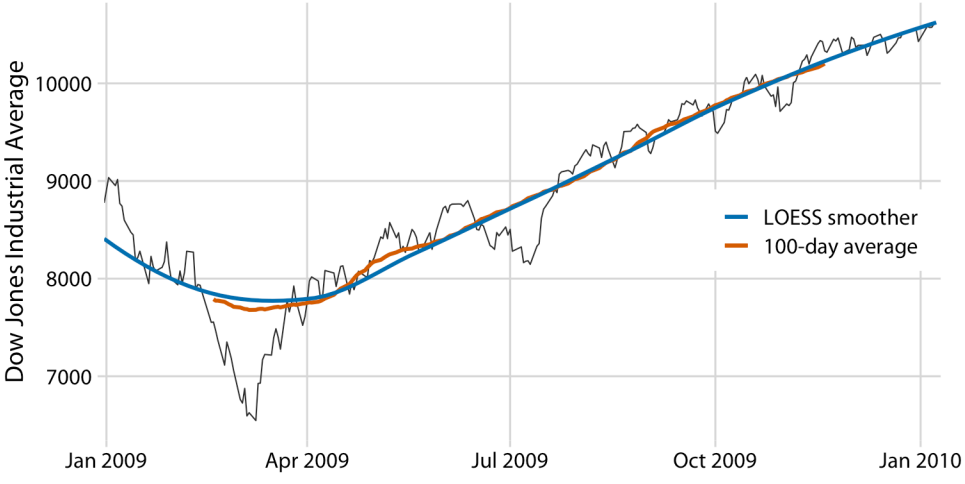


Figure 14-3. Comparison of LOESS fit to 100-day moving average for the Dow Jones data of Figure 14-2. The overall trend shown by the LOESS smooth is nearly identical to the 100-day moving average, but the LOESS curve is much smoother and it extends to the entire range of the data. Data source: Yahoo! Finance.

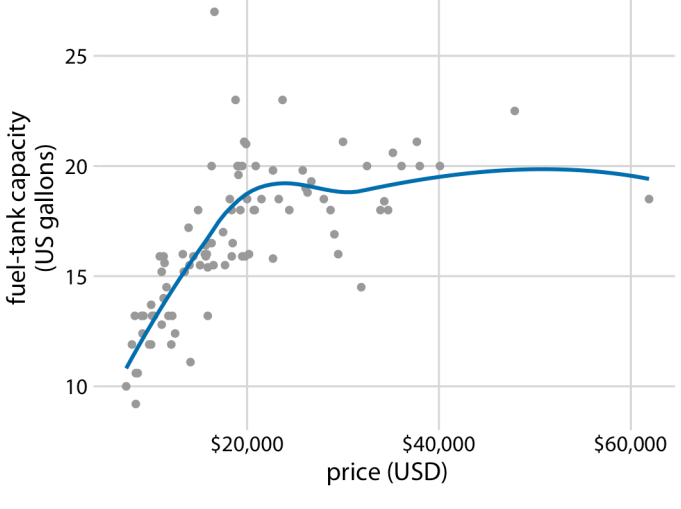


Figure 14-4. Fuel-tank capacity versus price of 93 cars released for the 1993 model year. Each dot corresponds to one car. The solid line represents a LOESS smooth of the data. We see that fuel-tank capacity increases approximately linearly with price, up to a price of approximately $20,000, and then it levels off. Data source: Robin H. Lock, St. Lawrence University.

LOESS is a very popular smoothing approach because it tends to produce results that look right to the human eye. However, it requires the fitting of many separate regression models. This makes it slow for large datasets, even on modern computing equipment.

As a faster alternative to LOESS, we can use spline models. A *spline* is a piecewise polynomial function that is highly flexible yet always looks smooth. When working with splines, we will encounter the term *knot.* The knots in a spline are the endpoints of the individual spline segments. If we fit a spline with k segments, we need to specify $k + 1$ knots. While spline fitting is computationally efficient, in particular if the number of knots is not too large, splines have their own downsides. Most importantly, there is a bewildering array of different types of splines, including cubic splines, B-splines, thin-plate splines, Gaussian process splines, and many others, and which one to pick may not be obvious. The specific choice of the type of spline and number of knots used can result in widely different smoothing functions for the same data (Figure 14-5).

Most data visualization software will provide smoothing features, likely implemented as either a type of local regression (such as LOESS) or a type of spline. The smoothing method may be referred to as a *generalized additive model* (GAM), which is a superset of all these types of smoothers. It is important to be aware that the output of the smoothing feature is dependent on the specific GAM model that is fit. Unless you try out a number of different choices you may never realize to what extent the results you see depend on the specific default choices made by your statistical software.



Be careful when interpreting the results from a smoothing function. The same dataset can be smoothed in many different ways.

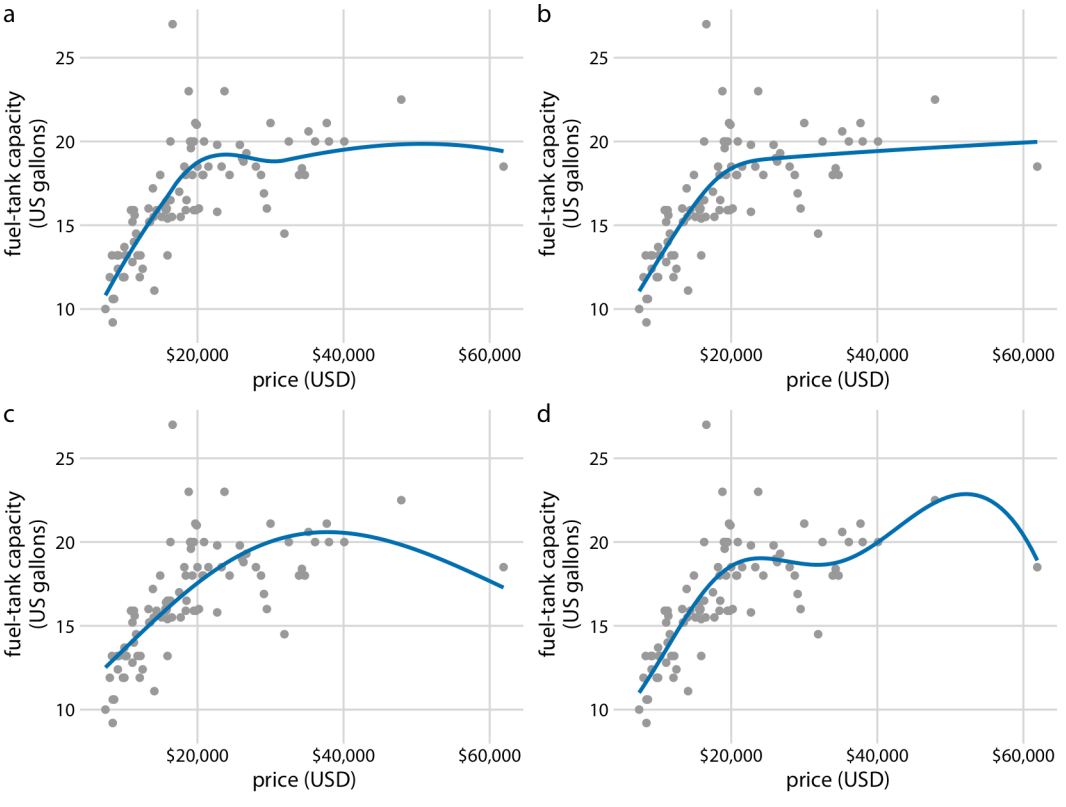


Figure 14-5. Different smoothing models display widely different behaviors, in particular near the boundaries of the data. (a) LOESS smoother, as in Figure 14-4. (b) Cubic regression splines with 5 knots. (c) Thin-plate regression spline with 3 knots. (d) Gaussian process spline with 6 knots. Data source: Robin H. Lock, St. Lawrence University.

Showing Trends with a Defined Functional Form

As we can see in Figure 14-5, the behavior of general-purpose smoothers can be somewhat unpredictable for any given dataset. These smoothers also do not provide parameter estimates that have a meaningful interpretation. Therefore, whenever possible, it is preferable to fit a curve with a specific functional form that is appropriate for the data and that uses parameters with clear meaning.

For the fuel-tank data, we need a curve that initially rises linearly but then levels off at a constant value. The function $y = A - B \exp(-mx)$ may fit that bill. Here, A, B, and m are the constants we adjust to fit the curve to the data. The function is approximately linear for small x, with $y \approx A - B + Bmx$; it approaches a constant value for large x, $y \approx A$, and it is strictly increasing for all values of x. Figure 14-6 shows that this

equation fits the data at least as well as any of the smoothers we considered previously (Figure 14-5).

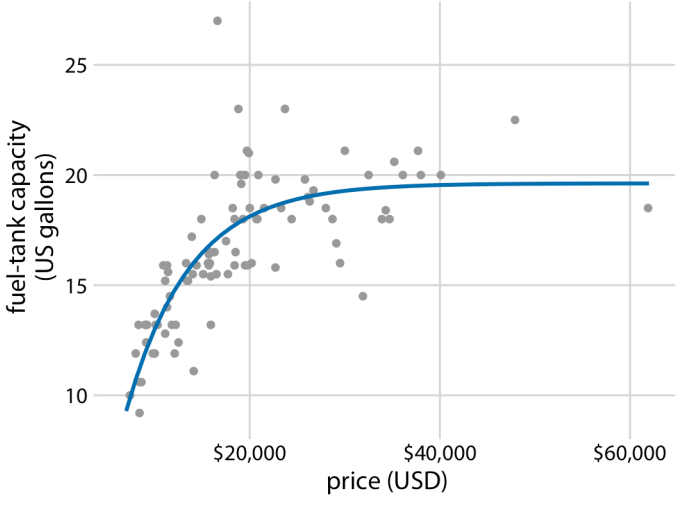


Figure 14-6. Fuel-tank data represented with an explicit analytical model. The solid line corresponds to a least-squares fit of the formula y = A – B *exp(*–mx*) to the data. Fitted parameters are* A = *19.6,* B = *29.2,* m = *0.00015. Data source: Robin H. Lock, St. Lawrence University.*

A functional form that is applicable in many different contexts is the simple straight line, $y = A + mx$. Approximately linear relationships between two variables are surprisingly common in real-world datasets. For example, in Chapter 12, I discussed the relationship between head length and body mass in blue jays. This relationship is approximately linear, for both female and male birds, and drawing linear trend lines on top of the points in a scatterplot helps the reader perceive the trends (Figure 14-7).

When the data displays a nonlinear relationship, we need to guess what an appropriate functional form might be. In this case, we can assess the accuracy of our guess by transforming the axes in such a way that a linear relationship emerges. To demonstrate this principle, let's return to the monthly submissions to the preprint server bioRxiv, discussed in Chapter 12. If the increase in submissions in each month is proportional to the number of submissions in the previous month—i.e., if submissions grow by a fixed percentage each month—then the resulting curve is exponential. This assumption seems to be met for the bioRxiv data, because a curve with exponential form, $y = A \exp(mx)$, fits the bioRxiv submission data well (Figure 14-8).

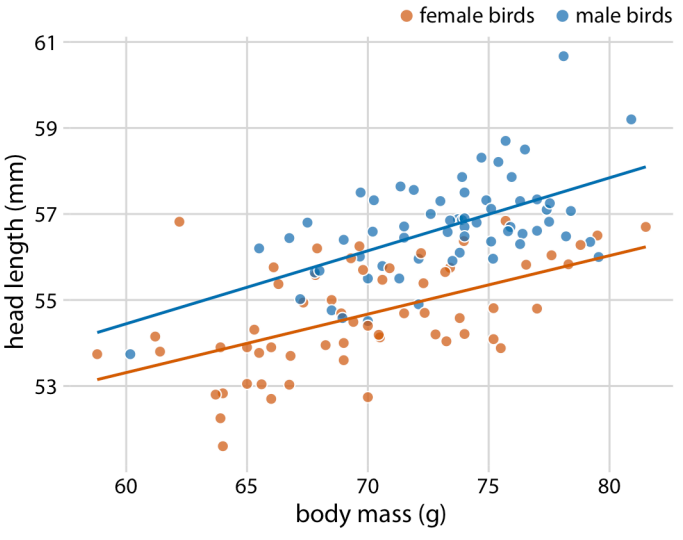


Figure 14-7. Head length versus body mass for 123 blue jays. The birds' sex is indicated by color. This figure is equivalent to Figure 12-2, except that now we have drawn linear trend lines on top of the individual data points. Data source: Keith Tarvin, Oberlin College.

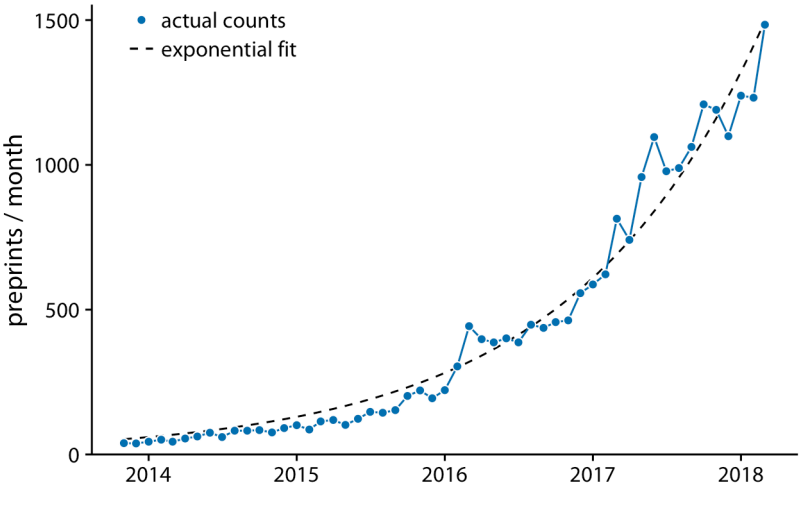


Figure 14-8. Monthly submissions to the preprint server bioRxiv. The solid blue line represents the actual monthly preprint counts and the dashed black line represents an exponential fit to the data, $y = 60 \exp[0.77(x - 2014)]$*. Data source: Jordan Anaya, http://www.prepubmed.org/.*

If the original curve is exponential, $y = A \exp(mx)$, then a log-transformation of the y values will turn it into a linear relationship, $\log(y) = \log(A) + mx$. Therefore, plotting the data with log-transformed y values (or equivalently, with a logarithmic y axis) and looking for a linear relationship is a good way of determining whether a dataset exhibits exponential growth. For the bioRxiv submission numbers, we indeed obtain a linear relationship when using a logarithmic y axis (Figure 14-9).

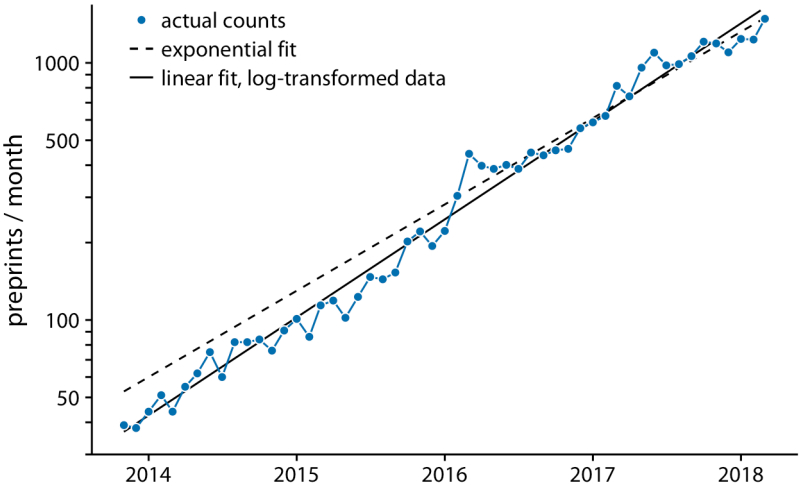


Figure 14-9. Monthly submissions to the preprint server bioRxiv, shown on a log scale. The solid blue line represents the actual monthly preprint counts, the dashed black line represents the exponential fit from Figure 14-8, and the solid black line represents a linear fit to log-transformed data, corresponding to y = *43 exp[0.88*(x – *2014)]. Data source: Jordan Anaya, http://www.prepubmed.org/.*

In Figure 14-9, in addition to the actual submission counts, I am also showing the exponential fit from Figure 14-8 and a linear fit to the log-transformed data. These two fits are similar but not identical. In particular, the slope of the dashed line seems somewhat off. The line systematically falls above the individual data points for half the time series. This is a common problem with exponential fits: the square deviations from the data points to the fitted curve are so much larger for the largest data values than for the smallest data values that the deviations of the smallest data values contribute little to the overall sum of squares that the fit minimizes. As a result, the fitted line systematically overshoots or undershoots the smallest data values. For this reason, I generally advise to avoid exponential fits and instead use linear fits on log-transformed data.



It is usually better to fit a straight line to transformed data than to fit a nonlinear curve to untransformed data.

A plot such as Figure 14-9 is commonly referred to as *log–linear*, since the y axis is logarithmic and the x axis is linear. Other plots we may encounter include *log–log*, where both the y and the x axis are logarithmic, and *linear–log*, where y is linear and x is logarithmic. In a log–log plot, power laws of the form $y \sim x^{\alpha}$ appear as straight lines (see Figure 8-7 for an example), and in a linear–log plot, logarithmic relationships of the form $y \sim \log(x)$ appear as straight lines. Other functional forms can be turned into linear relationships with more specialized coordinate transformations, but these three (log–linear, log–log, linear–log) cover a wide range of real-world applications.

Detrending and Time-Series Decomposition

For any time series with a prominent long-term trend, it may be useful to remove this trend to specifically highlight any notable deviations. This technique is called *detrending*, and I will demonstrate it here with house prices. In the US, the mortgage lender Freddie Mac publishes a monthly index called the *Freddie Mac House Price Index* that tracks the change in housing prices over time. The index attempts to capture the state of the entire house market in a given region, such that an increase in the index by, for example, 10% can be interpreted as an average house price increase of 10% in the respective market. The index is arbitrarily set to a value of 100 in December 2000.

Over long periods of time, house prices tend to display consistent annual growth, approximately in line with inflation. However, overlaid on top of this trend are housing bubbles that lead to severe boom and bust cycles. Figure 14-10 shows the actual house price index and its long-term trend for four select US states. We see that between 1980 and 2017, California underwent two bubbles, one in 1990 and one in the mid-2000s. During the same period, Nevada experienced only one bubble, in the mid-2000s, and house prices in Texas and West Virginia closely followed their long-term trends the entire time. Because house prices tend to grow in percent increments, i.e., exponentially, I have chosen a logarithmic y axis in Figure 14-10. The straight lines correspond to a 4.7% annual price increase in California and a 2.8% annual price increase each in Nevada, Texas, and West Virginia.

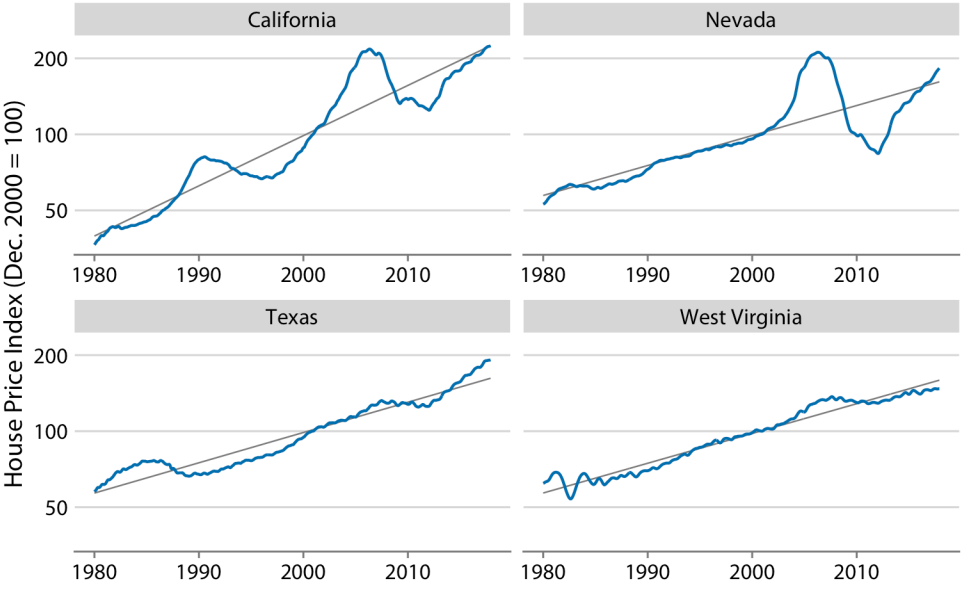


Figure 14-10. Freddie Mac House Price Index from 1980 through 2017, for four selected states (California, Nevada, Texas, and West Virginia). The House Price Index is a unitless number that tracks relative house prices in the chosen geographic region over time. The index is scaled arbitrarily such that it equals 100 in December of the year 2000. The blue lines show the monthly fluctuations in the index and the straight gray lines show the long-term price trends in the respective states. Note that the y *axes are logarithmic, so that the straight gray lines represent consistent exponential growth. Data source: Freddie Mac House Prices Index.*

We *detrend* housing prices by dividing the actual price index at each time point by the respective value in the long-term trend. Visually, this division will look like we are subtracting the gray lines from the blue lines in Figure 14-10, because a division of the untransformed values is equivalent to a subtraction of the log-transformed values. The resulting detrended house prices show the housing bubbles more clearly (Figure 14-11), as the detrending emphasizes the unexpected movements in a time series. For example, in the original time series, the decline in home prices in California from 1990 to about 1998 looks modest (Figure 14-10). However, during that same time period, on the basis of the long-term trend we would have expected prices to rise. Relative to the expected rise the drop in prices was substantial, amounting to 25% at the lowest point (Figure 14-11).

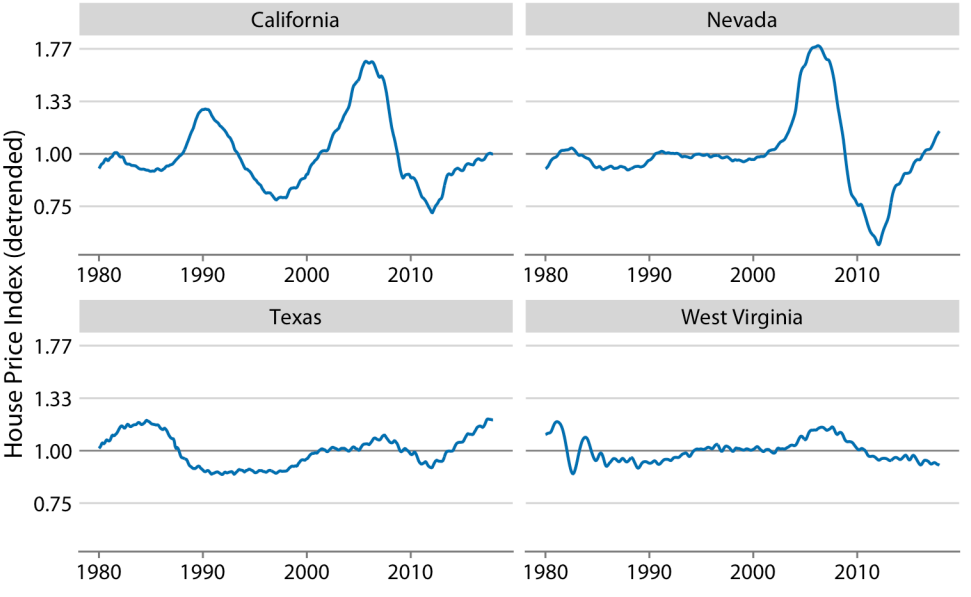


Figure 14-11. Detrended version of the Freddie Mac House Price Index shown in Figure 14-10. The detrended index was calculated by dividing the actual index (blue lines in Figure 14-10) by the expected value based on the long-term trend (straight gray lines in Figure 14-10). This visualization shows that California experienced two housing bubbles, around 1990 and in the mid-2000s, identifiable from a rapid rise and subsequent decline in the actual housing prices relative to what would have been expected from the long-term trend. Similarly, Nevada experienced one housing bubble, in the mid-2000s, and neither Texas nor West Virginia experienced much of a bubble at all. Data source: Freddie Mac House Prices Index.

Beyond simple detrending, we can also separate a time series into multiple distinct components, such that their sum recovers the original time series. In general, in addition to a long-term trend, there are three distinct components that may shape a time series. First, there is random noise, which causes small, erratic movements up and down. This noise is visible in all the time series shown in this chapter, but maybe the most in Figure 14-9. Second, there can be unique external events that leave their mark in the time series, such as the distinct housing bubbles seen in Figure 14-10. Third, there can be cyclical variations. For example, outside temperatures show daily cyclical variations. The highest temperatures are reached in the early afternoon and the lowest temperatures in the early morning. Outside temperatures also show yearly cyclical variations. They tend to rise in the spring, reach their maximum in the summer, and then decline in the fall and reach their minimum in the winter (Figure 3-2).

To demonstrate the concept of distinct time-series components, I will here decompose the Keeling curve, which shows changes in CO_2 abundance over time (Figure 14-12). Since 1958, CO_2 abundance has been continuously monitored at the Mauna Loa Observatory in Hawaii, initially under the direction of Charles Keeling.

CO_2 is measured in parts per million (ppm). We see a long-term increase in CO_2 abundance that is slightly faster than linear, from below 325 ppm in the 1960s to above 400 in the second decade of the 21st century (Figure 14-12). CO_2 abundance also fluctuates annually, following a consistent up-and-down pattern overlaid on top of the overall increase. The annual fluctuations are driven by plant growth in the northern hemisphere. Plants consume CO_2 during photosynthesis. Because most of the globe's land masses are located in the northern hemisphere, and plant growth is most active in the spring and summer, we see an annual global decline in atmospheric CO_2 that coincides with the summer months in the northern hemisphere.

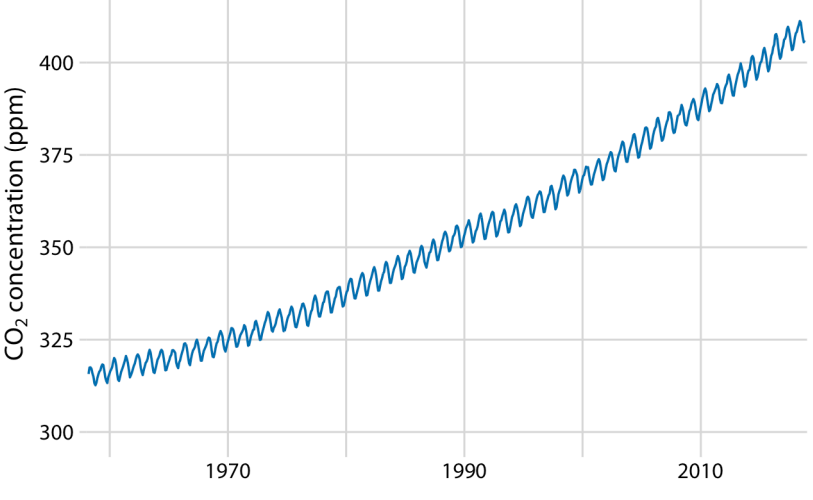


Figure 14-12. The Keeling curve. The Keeling curve shows the change of CO_2 abundance in the atmosphere over time. Shown here are monthly average CO_2 readings, expressed in parts per million (ppm). The CO_2 readings fluctuate annually with the seasons but show a consistent long-term trend of increase. Data source: Dr. Pieter Tans, NOAA/ESRL, and Dr. Ralph Keeling, Scripps Institution of Oceanography.

We can decompose the Keeling curve into its long-term trend, seasonal fluctuations, and remainder (Figure 14-13). The specific method I am using here is called *seasonal decomposition of time series by LOESS* (STL) [Cleveland et al. 1990], but there are many other methods that achieve similar goals.

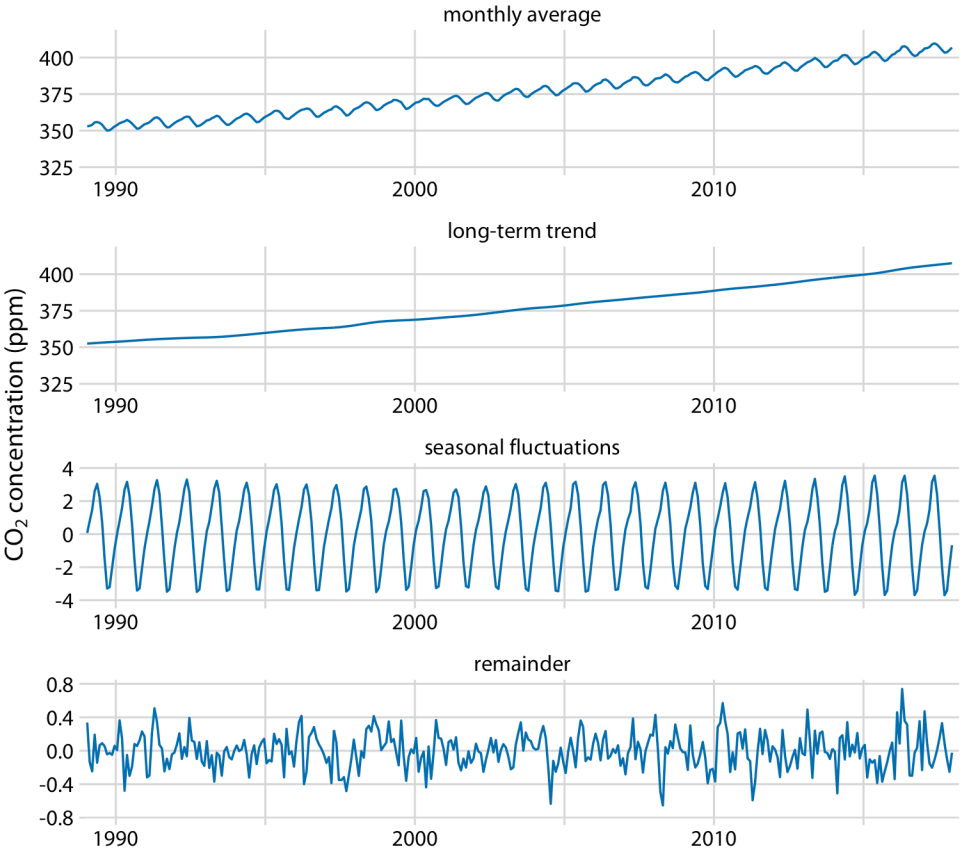


Figure 14-13. Time-series decomposition of the Keeling curve, showing the monthly average (as in Figure 14-12), the long-term trend, seasonal fluctuations, and the remainder. The remainder is the difference between the actual readings and the sum of the long-term trend and the seasonal fluctuations, and it represents random noise. I have zoomed into the most recent 30 years of data to emphasize the shape of the annual fluctuations. Data source: Dr. Pieter Tans, NOAA/ESRL, and Dr. Ralph Keeling, Scripps Institution of Oceanography.

The decomposition shows that over the last three decades, CO_2 abundance has increased by over 50 ppm. By comparison, seasonal fluctuations amount to less than 8 ppm (they never cause an increase or a decrease of more than 4 ppm relative to the long-term trend), and the remainder amounts to less than 1.6 ppm (Figure 14-13). The remainder is the difference between the actual readings and the sum of the long-term trend and the seasonal fluctuations, and here it corresponds to random noise in the monthly CO_2 readings. More generally, however, the remainder could also

capture unique external events. For example, if a massive volcano eruption released substantial amounts of CO_2, such an event might be visible as a sudden spike in the remainder. Figure 14-13 shows that no such unique external events have had a major effect on the Keeling curve in recent decades.